

Model Performance Measurements

1. Introduction

The goal of the exercise is to change different parameters to observe their impact on the AUC and R2 score.

2. Changing the Kernel

SVM Kernel	AUC value
Linear	0.79
rbf	0.74
poly	0.85

I changed the SVM kernels from linear to RBF and to poly to compare their AUC values.

Selecting an appropriate kernel can improve model performance. In this case, the polynomial kernel performed best because it achieved the highest AUC.

3. Increasing and Decreasing Noise

Noise	AUC value
200	0.79
50	0.91
1000	0.53

I increased and decreased the number of noise features to observe their effect on the model performance. If the number of noise features decreases, the model

performs better (0.91) because it has access to more relevant information. On the other hand, adding more noise features reduces model performance (0.53) because the model is affected by irrelevant variables that make the classification task more difficult.

4. Effect of Changing Prediction Values on R2 Score

To observe the effect of prediction accuracy on the R2 score, I changed the predicted values.

Y prediction	Y true	R2 score
2.5, 0.0, 2, 8	3, -0.5, 2, 7	0.95
3, -0.5, 2, 7	3, -0.5, 2, 7	1
1, 4, -0.3, -5	3, -0.5, 2, 7	-4.95

To see the impact of prediction accuracy on R2 performance, I changed the values in `y_pred`. It is possible to observe that if the predicted values are equal to the true values, the R2 score is one, indicating a perfect prediction. When the prediction is slightly different, the prediction performance decreases but remains good (0.95), while if I replace them with values that are completely different from the true values, the R2 score becomes negative, showing poor predictive performance.

5. Conclusion

In conclusion, changing the parameters of the model, such as the SVM kernel and the number of noise features, can have a significant impact on the AUC score. In addition, changing the predicted values used to calculate the R2 score shows how prediction accuracy can affect regression performance. Overall, the

results show that the model performance is strongly influenced by the parameters and the quality of the data.